

COMPUTATIONAL ANALYSIS OF COMMUNICATION

CRIME FRAMING

18.05.2026

Robin-Kiara Braun & Marmee Pandya

Annotation Instruction

- Our dimension focuses on the **Immigration and Crime Frame**, operationalised as three categories following the framing literature:
 - **CRIME_FRAME_NEG**
 - **CRIME_FRAME_POS**
 - **NO_CRIME_FRAME**
- Prompt includes a role description, label definitions with explicit decision rules, boundary cases, and a reminder at the end.

Crime Frame Negative

- Direct perpetrator: suspect, accused, charged. Court confirmation not required
- Terror, extremism: connected to or legitimising violent acts
- Security risk framing: politician or institution frames group as threat to Germany
- Generalised threat: group brings Gewalt, Chaos or instability to Germany
- Domestic authoritarian control: coercive behaviour in Germany
- Crime abroad: only if explicit Germany connection exists

Crime Frame Positive

- Prevention and social work, deradicalisation, integration programmes
- Explicit differentiation, "kein Generalverdacht", "not all members"
- Crime explained through social factors, not the group itself
- NOT: arrests, deportations, police measures against group members then those are CRIME_FRAME_NEG

No Crime Frame

- War reporting and Geopolitical conflicts
- Fictional or Cultural depictions of crime
- Propaganda and debunked narratives
- Group is named as a victim of violence, racism, discrimination or state repression
- Violence/Crime happening outside Germany with no explicit Germany connection
- Crime attributed to social conditions (poverty, exclusion) not the group itself
- Demonstrations and protests without explicit link to crime or security measures

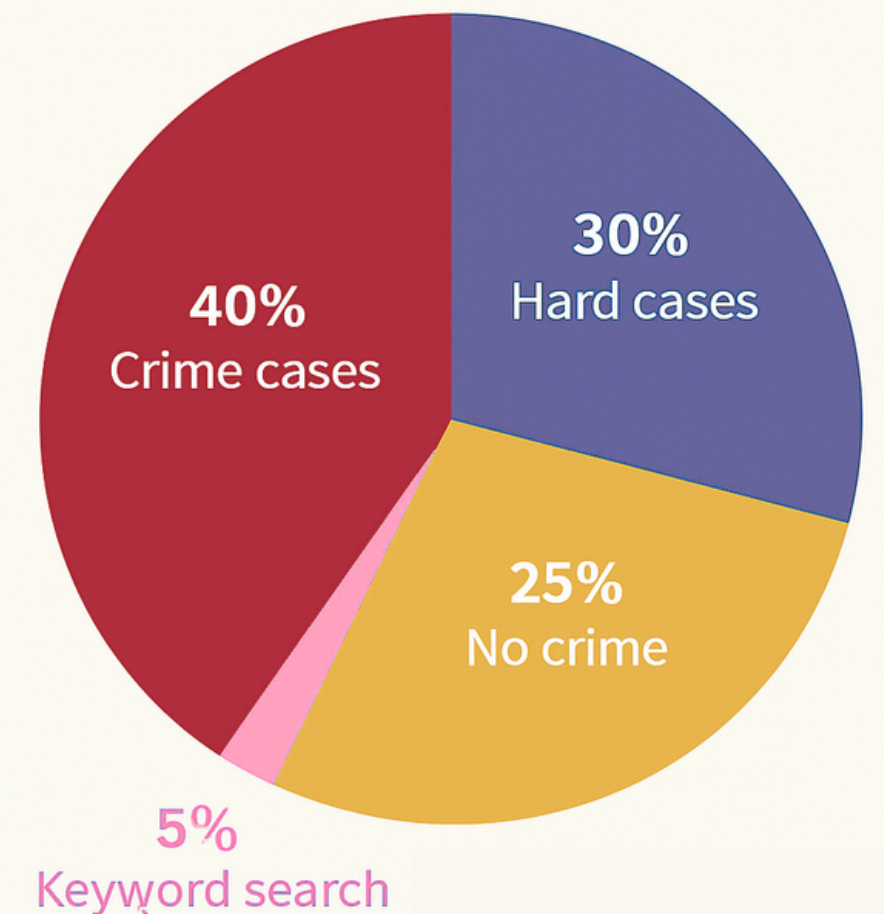
Dataset

Sample of 2000

- run 10x on each paragraph with temperature 0.7
 - 266 (13.3%) cases less than 80% agreement
- manually labeled hardcases

Sample of 500

- oversample crime cases
- 40% crime cases, 30% Hard cases, 25% no crime and 5 % keyword search
 - "theft," "terrorism," "security," "attack," "prevention"



Human Annotation

Inter-Rater Agreement & Krippendorff's Alpha

- Manually annotated 500 paragraphs from the sample in two blocks:

	Block A (N=250)	Block B (N=250)
Rows with agreement	192 (76.8%)	236 (94.4%)
Rows without agreement	58 (23.2%)	14 (5.6%)
Krippendorff's alpha	0.560	0.786

- Final human label distribution across 500 rows:
 - 69% no crime, 30% negative crime, 1% positive crime

Model performance

- We tested three annotation conditions using minstral-3b:

Approach	Kappa	F1
zero shot (baseline)	0.415	0.563
few-shot	0.638	0.712
few-shot + chain-of-thought	0.472	0.574
Validation (fresh 250)	0.702	0.849

Error Analysis

- Few-Shot learning precision for CRIME_FRAME_NEG 0.84, recall 0.74 model is precise but **misses some indirect or generalised threat framing**
- CRIME_FRAME_POS **too rare** in the data to evaluate reliably
- Validation on fresh 250: kappa 0.702, above the first 250 result (0.588) pipeline generalises to unseen data

Next Steps

- Annotate more data
 - sample with 10% crime cases
- drop positive crime label?
- report